

ASHUTOSH ROY

AI Engineer | LLM Systems, RAG & Agentic Workflows

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SUMMARY

AI Engineer with production experience in LLM-powered agentic systems, RAG pipelines, and multimodal document AI. Published researcher (RECCAP 2026, IIT Palakkad; co-author IIT Delhi) with expertise in tool-calling agents via **LangGraph** and **Model Context Protocol (MCP)**, hybrid retrieval over unstructured data (documents, OCR, schemas), and deploying **FastAPI** backends where correctness and low latency are non-negotiable. Focused on safety-aware AI where abstention and human-in-the-loop are first-class requirements.

EXPERIENCE

Research Intern

Jan 2025 – Jun 2025

Indian Institute of Technology (IIT) Delhi

New Delhi, India

- Built **layout-aware document AI pipelines** (Tesseract, Marker, Gemini OCR) for parsing degraded historical records, schematics, and scanned engineering documents — dynamically selecting extraction strategy per document class.
- Designed **hybrid retrieval pipelines** (FAISS/Pinecone dense + Apache Solr sparse) enabling semantic and keyword search over heterogeneous document corpora; reduced RAG latency by **35%**.
- Deployed **FastAPI** services for real-time LLM inference with function calling; integrated with frontend for sub-second multi-modal retrieval at production scale.
- Co-authored **SARCH: Multimodal Search for Archaeological Archives** — submitted for peer review. arxiv.org/abs/2511.05667

AI/ML Engineer Intern

Oct 2024 – Nov 2024

MitoVoid AI

Remote

- Fine-tuned **Mistral/LLaMA** via QLoRA for healthcare domain; built conversational AI agents with autonomous **tool-calling** for medical record summarization and intent routing.
- Reduced patient query resolution time by **~40%** through optimized agentic intent routing; instrumented API observability for latency and failure tracking.
- Built end-to-end ML pipelines (ingestion → preprocessing → inference → deployment) and OCR modules for structured extraction from clinical reports.

Software Engineer Intern

Nov 2024 – Dec 2024

LeanTactics Solutions Pvt. Ltd.

Remote

- Engineered **FastAPI** backend services integrating ML model outputs into production systems; profiled API bottlenecks achieving **25–35% latency reduction** under production load.

PROJECTS

Shopping Assistant Agent (LangGraph & MCP) | Python, LangGraph, MCP, Claude 3.5, FastAPI

Feb 2026 – May 2026

- Engineered a **multi-agent system** using LangGraph to coordinate specialized agents (search, comparison, recommendation) with self-reflection and retry loops for correctness under uncertainty.
- Implemented custom **Model Context Protocol (MCP)** servers for secure, real-time access to product APIs and external data sources — demonstrating deep MCP integration and tool routing.

Skills for Agents Framework | Open-source, Python, AI Agents, SIP

Apr 2026 – May 2026

- Designed a production-ready ecosystem of composable instruction sets allowing AI agents to handle concurrent tasks without prompt collision — directly applicable to multi-step industrial workflows.
- Architected the **Skills Interoperability Protocol (SIP)** for domain-separated agent capabilities, enabling safe sequential and parallel pipeline behaviors.

Exam-Helper Agentic RAG System | Python, LLaMA 3, LangGraph, FAISS, Pinecone

Dec 2024 – Apr 2025

- Built an LLM-powered QA agent on **LLaMA 3** with multi-step reasoning over PDFs, images, and transcripts; FAISS/Pinecone dense retrieval with semantic chunking for hallucination-minimized responses.

Speech Emotion & Stress Detection | PyTorch, Wav2Vec2, BiLSTM | RECCAP 2026, IIT Palakkad

Jan 2026 – May 2026

- First-authored paper on self-supervised Wav2Vec2 + BiLSTM multitask architecture; **85.9% accuracy**, RMSE **0.1268** — peer-reviewed and accepted at RECCAP 2026.

TECHNICAL SKILLS

LLMs & Agents: LangChain, LangGraph, Model Context Protocol (MCP), Tool Use / Function Calling, Multi-Agent Orchestration, Prompt Engineering, RAG, Guardrails & Hallucination Control

Retrieval & Data: FAISS, Pinecone, Apache Solr, Hybrid Search, Layout-Aware OCR, Document Parsing (Tesseract, Marker, Gemini OCR)

Backend & Deployment: FastAPI, Docker, REST APIs, Git, GitHub Actions, Linux, Observability (Langfuse)

ML & Research: PyTorch, Hugging Face, Transformer Fine-Tuning (QLoRA/LoRA), Scikit-learn, Pandas, NumPy

Languages: Python, C++, SQL

EDUCATION

Chhattisgarh Swami Vivekananda Technical University (CSVTU)

2022 – 2026

B.Tech (Hons) in Computer Science and Engineering (Artificial Intelligence)

Bhilai, Chhattisgarh

St. Columbus School

2020 – 2021

Higher Secondary (Class XII - Science PCMB) - 87%

Faridabad, Haryana